MIL-S-7809

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Superseding
AN-QQ-S-676
17 June 1941

MILITARY SPECIFICATION

STEEL, LOW ALLOY, HIGH STRENGTH, PLATE, SHEET, AND STRIP
(FOR AIRCRAFT APPLICATIONS)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 Scope.- This specification presents requirements for low-alloy high-strength steel plate, sheet, and strip products.

1.2 Classification.- Either hot-rolled or cold-rolled material may be furnished in the as-rolled, annealed, normalized, or stress-relieved condition. Drawing quality may be furnished. Surface scale and oxide shall be removed by pickling.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.-

Federal

QQ-M-151	Metals; General Specification for Inspection of
P-S-661	Solvent; Dry-Cleaning

Military

MIL-W-6858	Welding, Low Carbon Steels, Austenitic Steel, and Nickel Alloys, Spot and Seam

Air Force-Navy Aeronautical

AN-O-6	Oil; General Purpose, Low Temperature Lubricating

2.1.2 Other Publications.-

Military Standards (Book)

MIL-STD-129	Marking of Shipments

2.1.3 Drawings.-

Air Force-Navy Aeronautical Standard Drawings

AND10355	Tolerances - Carbon Steel Sheet and Strip

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.-

3.1.1 Chemical Composition.- Material shall conform to the following requirements:

Element	Analysis (maximum) percent by weight
Carbon	0.20
Manganese	1.25
Sulfur	0.050
Phosphorus	0.150
Silicon	0.90

Other elements such as nickel, chromium, copper, molybdenum, aluminum, titanium, and zirconium may be used singly or in combination with the elements listed above to achieve the required mechanical properties.

3.1.1.1 The manufacturer shall furnish a ladle analysis of each heat of steel from which material is furnished on the contract.

3.2 Defects.- All plate, sheet, and strip shall be smooth, free from seams, laminations, blisters, scale, oxide, and other defects which exceed in depth the minimum thickness tolerance, and shall have a good workmanlike finish. Spot grinding not to exceed the minimum thickness tolerance is permissible.

3.3 Mechanical Properties.- The mechanical properties shall be as follows:

	Thickness 1/2 Inch and Under	Thickness Over 1/2 to 1-1/2 Inches Inclusive
Yield point (min) psi	50,000	47,000
Tensile strength, (min) psi	70,000	67,000
Elongation in 2 inches (min) percent	1/ 22	- -
Elongation in 8 inches (min) percent (0.180 inch and heavier)	$\left(\frac{1,500,000}{\text{Tensile Strength, PSI}}\right)$	$\left(\frac{1,500,000}{\text{Tensile Strength, PSI}}\right)$

1/ For cold-reduced sheets 0.070 inch or less in thickness, the percent elongation in 2 inches is 20 percent.

(a) For material over 3/4 inch in thickness, a deduction from the percentage of elongation in 8 in. of 0.25 percent shall be made for each increase of 1/16 inch of the specified thickness above 3/4 inch.

(b) For material under 5/16 inch in thickness, a deduction from the percentage of elongation in 8 inches of 2.00 percent shall be made for each decrease of 1/32 inch of the specified thickness below 5/16 inch.

(c) The minimum yield point and tensile strength requirements for hot-rolled products are reduced by 5,000 psi when material is specified in the annealed condition.

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3.3.1 Bending.- Materials not over 3/4 inch in thickness shall withstand cold bending with the fold of the bend transverse to the direction of rolling, without cracking, through an angle of 180 degrees, over a diameter equal to the bend factor "N" (as indicated below) times the thickness of the plate, sheet, or strip:

Thickness of Material (Inch)	Bend Factors N
0.500 and less	1
0.501 to 1.000 incl.	2
1.001 to 2.000 incl.	3

3.4 Weldability.- The material shall possess good weldability, such that sound weld deposits can be obtained by ordinary fusion welding methods in materials 0.062 inch or more in thickness. Upon cooling in air after arc welding (without preheat or postheat) the maximum hardness of the heat-affected zone shall not exceed the average hardness of the base metal by more than 20 percent, as computed on the Rockwell "B" scale.

3.4.1 The material shall be suitable for spot welding in accordance with Specification MIL-W-6858.

3.5 Tolerances.- Dimensional tolerances shall be in accordance with the limits of acceptable variation as indicated by Drawing AND10355.

3.6 Identification of Product.- Each flat plate, sheet, or strip shall be legibly marked on one side only, with rows of recurring symbols which will identify the product. The following information shall be included:

MIL-S-7809
Manufacturer's Name or Trade-Mark
Nominal Thickness (optional)*
Commercial Designation (optional)

* If the nominal thickness is not included in the continuous marking indicated above, it shall be indicated on the opposite corners of each sheet.

The number of rows to be marked on each sheet shall be determined by the width of the sheet as indicated below:

Width	No. of Rows of Marking
1/2 to 3 inches	1
3 to 7 inches	2
Over 7 inches	1 for every 3 inches

3.6.1 Marking Fluid.- Markings shall be applied with a marking fluid which is not affected by water or solvent conforming to Specification P-S-661 and shall not rub off or be smeared by contact incident to normal handling and shipment.

3.6.2 Marking of Strip Furnished in Coils.- Identification marking as specified herein shall be applied to the outside end of the strip, and when practicable, to the inside end also.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.- All the tests required herein for the testing of steel are classified as Inspection tests, for which necessary sampling techniques and methods of testing are specified in this section. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.2 Examination of Product.- Sufficient spot checks shall be made to assure compliance with the surface condition, marking, dimensional, and defect requirements.

4.3 Chemical Analysis.-

4.3.1 Sampling.- One or more samples for check chemical analysis shall be selected to represent each heat of steel from which material is furnished on contract.

4.3.1.1 When the producer's method of composition control is acceptable, samples for check chemical analysis may be waived at the discretion of the Government Inspector.

4.3.1.2 Additional Samples.- Where the material is not identifiable as to heat and method of manufacture, or where the identity of any portion of the shipment is obscure in any respect, the Inspector shall select the necessary additional samples for check chemical analysis to determine conformance of all portions of the shipment to this specification.

4.3.2 Method.- Specimens shall be prepared in accordance with Specification QQ-M-151 and shall be tested by wet chemical, spectrographic, or other analytical methods. In the event of dispute, analysis shall be by wet chemical methods.

4.4 Tensile Test.-

4.4.1 Sampling.- Samples shall be taken from each of two plates, sheets, or strips from the same heat, processed under the same conditions, of the same thickness, and submitted for acceptance at one time. When several thicknesses of sheet or plate are rolled from the same heat, two Tensile and two Bend test samples each shall be taken from the thinnest and from the thickest material rolled. In the case of annealed materials, Tensile and Bend test samples shall be taken from each annealing charge represented.

4.4.2 Method.- Specimens conforming to types 2, 3, or 5 of Specification QQ-M-151 shall be machined from the samples and loaded to failure in accordance with the applicable requirements of Specification QQ-M-151. Yield strength, when required, shall be determined by the offset or extension-under-load methods.

4.5 Weldability Test.-

4.5.1 Sampling.- One pair of weldability test 6- by 8-inch plates shall be selected from the thinnest plate, sheet, or strip representative of each heat from which materials are furnished on an order.

4.5.2 Preparation of Specimen.- A butt joint not less than 6 inches in length shall be arc-welded between the two test plates, employing electrodes, conditions, and procedures selected in accordance with good commercial practice. No preheating or post-heating shall be applied.

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4.5.3 Method.- The weld beads shall be examined visually and/or at low magnification for evidence of weld defects. A section shall be made transverse to the weld line, polished to appropriate fineness, examined for porosity or cracking in or near the deposited metal, and a hardness traverse made to determine the maximum hardness in the heat-affected zones adjacent to the weld deposit as well as that of the base metal. The maximum increase in hardness due to welding heat shall be computed. The presence of weld defects attributable to the compositions or structure of the base metal shall be cause for rejection.

4.6 Bend Test.-

4.6.1 Sampling.- Samples shall be taken from each of two plates, sheets, or strips from the same heat, processed under the same conditions, of the same thickness, and submitted for acceptance at one time.

4.6.2 Method.- Specimens shall be bent cold either by pressure or by blows, except that in the case of dispute, tests shall be made by pressure. The axis of the bend shall be parallel to the direction of rolling.

4.7 Rejection.- Where failure of a sample or specimen is definitely ascribed to faulty material, the entire lot of material represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Application.- The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation.- A protective coating of oil conforming to Specification AN-O-6, or oil of equivalent protective properties, shall be applied to the surfaces of all products.

5.3 Packaging.- Products preserved as specified above shall be separated by size and condition when packed for shipment and shall be prepared in units convenient to handle at destination.

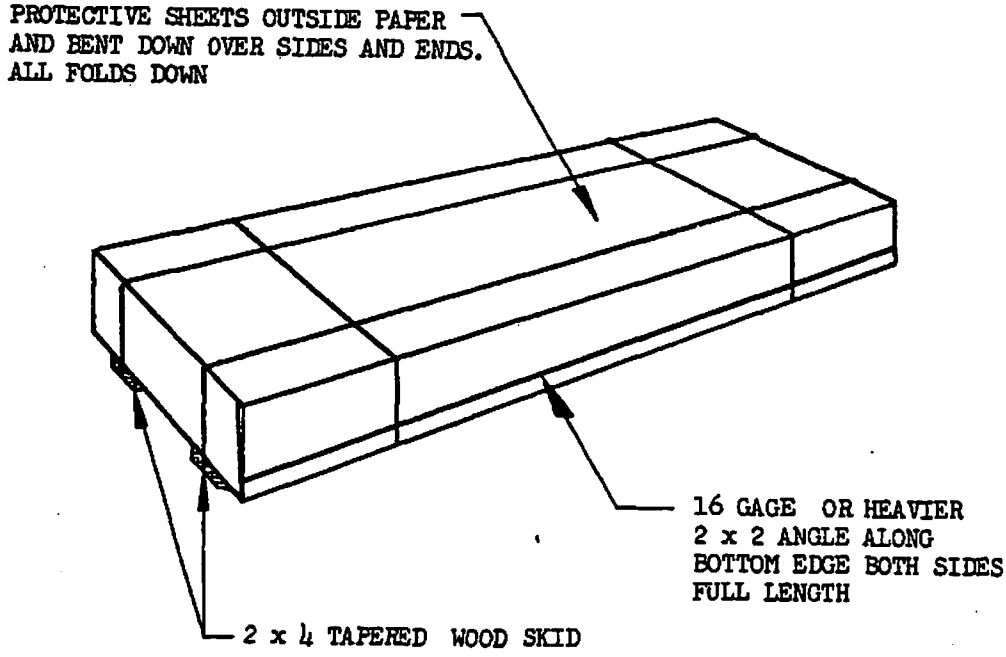
5.4 Packing.- Unless otherwise specified, material shall be packed for domestic shipment.

5.4.1 Domestic Packing.- Containers shall be so constructed as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery. Except as specified herein, the containers shall conform to the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment, and shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.4.2 Overseas Packing.- The gross weight of a fully packed shipping container shall not exceed approximately 1,000 pounds.

5.4.2.1 Flat Sheets and Plates.- Sheets shall be mounted on 2- by 4-inch skids, apped with a layer of grease-resistant paper, overwrapped with a layer of string-inforced water-resistant paper, and covered by bent waster sheets as indicated by gure 1. Tops, sides, and ends shall be one continuous piece. Lap-down on sides and ds shall cover the sides and ends completely. Bands or ties shall be not less than ¼ inch by 0.035 inch flat steel bands, tension tied, or 8-gage galvanized wire. Corner otectors shall be used under wire ties at all corners. For lifts up to 120 inches long, o lengthwise ties and skids shall be used for sheet widths 54 inches and under; three gthwise ties and skids for sheets over 54 inches. Two crosswise ties shall be used for gths of 72 inches and under; three crosswise ties for lengths over 72 inches.

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DIMENSIONS IN INCHES

FIGURE 1. Methods of Packaging Sheets.

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5.4.2.2 Coiled Sheet and Strip.- Coils shall be wrapped with grease-resistant paper, overwrapped with water-resistant paper, and then with thin steel waster sheets. Each coil shall be tension-tied with four or more transverse bands and one or more circumferential bands of 3/4- by 0.035-inch flat steel bands. Corner protectors shall be used under the bands.

5.5 Marking and Labeling.- Each shipping container shall be marked with the following information:

STEEL, LOW CARBON AND LOW ALLOY, SHEET, (PLATE OR STRIP)
Specification MIL-S-7809
Size and Thickness
Heat No.
Quantity
Contract or Order No.
Manufacturer's Name or Trade-Mark
Name of Contractor (if different from manufacturer)

5.5.1 In addition, each shipping container shall be marked in accordance with the requirements applicable to the individual Services, as specified in Standard MIL-STD-129.

6. NOTES

6.1 Intended Use.- The steel plate, sheet and strip procurable under this specification possesses fair forming properties and good weldabability and is suitable for general use in the manufacture of aircraft fittings and miscellaneous parts which are not heat-treated to obtain high mechanical properties.

6.2 Ordering Data.- Requisitions, contracts, and purchase orders should state the size and thickness of products to be furnished, and whether domestic or overseas packing is desired.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Navy - Bureau of Aeronautics
Air Force

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